

$$\begin{aligned}
 F_{n,qs} &= K_{p,n} n_s^T (e_p - e_{p,0}) \\
 &= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N}
 \end{aligned}$$